

Andre Arcaina

 andrearcaina  andrearcaina  andrearcaina.ca

Education

Toronto Metropolitan University (TMU, formerly Ryerson University)

Sept 2022 – Present

Bachelor of Science (Honours), Computer Science (Co-op)

Toronto, ON

Coursework: Data Structures, Algorithms, Databases, Computer Networks, Operating Systems, Software Engineering, Object Oriented Programming, Distributed Systems, Big Data Systems

Work Experience

Environment and Climate Change Canada (ECCC)

June 2025 – Present

Software Engineer Intern

North York, ON

- Identified lack of automated testing in Jenkins CI and built a Go-based end-to-end testing framework to validate component data flow across multiple test cases, reducing bug triage time in satellite data pipelines.
- Led modernization of mission-critical Java services by migrating 8 components to Java 17 and Jakarta EE 10, improving maintainability and supporting national weather data delivery systems used in maritime and Coast Guard operations.
- Eliminated decades of technical debt by rebuilding a legacy VB6 application from the ground up in Python/PyQt with a PostgreSQL backend, securely processing datasets exceeding 50,000 rows.
- Led the documentation and maintainability of a legacy C++ and Java system, authoring foundational architecture and UML diagrams to accelerate future feature development for a 3-person team.
- Refactored 3 legacy C++ services by removing deprecated exception patterns, modernizing error handling, and improving code clarity, reducing technical debt and improving long-term maintainability across production systems.

Undergraduate Science Society of TMU (USSTM)

Jan 2025 – Apr 2026

Backend Engineer

Toronto, ON

- Architected a campus vault backend for the undergraduate science student body by engineering a contract-first Go RESTful API across 32+ endpoints, leveraging SQLc for type safety and OpenAPI for strict schema enforcement.
- Led backend reliability initiatives by developing 37+ unit and integration tests, achieving 85%+ code coverage and preventing regressions in core business logic.
- Built AWS S3 image storage with presigned URLs to support secure, horizontally scalable before and after item image uploads.
- Developed asynchronous email processing using Redis-backed Asynq workers and AWS SES for reliable notification delivery.
- Collaborated with the frontend team to deliver scalable platform features aligned with organizational requirements.

DataKinetics

May 2024 – Aug 2024

Software Engineer Intern

Ottawa, ON

- Eliminated manual data inspection for the R&D team by developing a Spring Boot RESTful middleware API and web dashboard that automated COBOL-to-JSON parsing using an external Java library, documented end-to-end in Confluence.
- Presented the finalized tool to 12 R&D stakeholders, successfully demonstrating the new automated data-inspection workflow.
- Designed and executed 72 comprehensive Postman test cases for a core mainframe REST API, rigorously validating JSON responses, edge cases, and data integrity across 3 IBM Db2 databases.

Projects

Distributed Cloud-Native Trading Platform | Go, gRPC, Docker, Kubernetes, NATS, PostgreSQL, Redis

GitHub

- Architected a distributed paper trading platform composed of 8+ Go microservices, sustaining 1,600+ requests per second across 5,000 concurrent virtual users during load testing with a 0% error rate (P95: 42ms, Avg: 7ms).
- Led infrastructure deployment using Kubernetes and Helm, integrating system observability with Prometheus, Loki and Grafana.
- Reduced market data request latency from 800ms to 10ms by implementing Redis caching and request coalescing.

Distributed Key-Value Database | Go, Raft, BadgerDB

GitHub

- Developed a persistent distributed key-value store in Go backed by BadgerDB, implementing Raft leader election and replicated log coordination for fault-tolerant state replication across multiple node processes.

Skills

Languages: Go, Python, Java, C++, Rust, JavaScript, TypeScript, SQL, GraphQL, Bash

Frameworks: Flask, FastAPI, Spring Boot, Node.js, React.js, Next.js, TailwindCSS

Tools: GCP, AWS, Docker, Kubernetes, NATS, PostgreSQL, Redis, Linux, Git, Jenkins, GitHub Actions